



Combined effects of super absorbent polymers and microbial fertilizers Co-application on apple production in the Loess Plateau

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ABSTRACT

Super absorbent polymers (SAP) and microbial fertilizers (MF) are pivotal in ameliorating soil physicochemical properties, regulating plant growth, and enhancing fruit quality. However, the combined effects of their co-application on apple plants remain unclear. To address this, a two-year field experiment was conducted on the Loess Plateau to evaluate different co-application ratios of SAP and MF. Results showed that the combined application improved soil properties, with higher contents of available nitrogen and organic matter, better moisture retention and lower soluble salt content. These improvements were accompanied by significant increases in chlorophyll content, net photosynthetic rate (Pn), water use efficiency (WUE), hundred-leaf weight, and fruit yield. Fruit quality was also enhanced, as evidenced by higher single fruit weight, vitamin C (VC), and soluble solids contents in apple fruits. Comprehensive evaluation identified 180 g SAP combined with 750 g MF per plant (P3M3) as the optimal treatment. Transcriptomic analysis revealed that P3M3 upregulated genes involved in nitrogen metabolism and arginine biosynthesis, consistent with elevated free amino acid levels in the roots. 16S rRNA sequencing further showed that P3M3 restructured the bacterial communities of the rhizosphere and bulk soil without reducing their diversity, enriching Actinobacteriota and diazotrophic genera including Frankia and Bradyrhizobium, and enhancing the predicted potential for nitrogen fixation. Overall, our findings reveal the associations among soil properties, bacterial communities, and root metabolism under SAP and MF co-application, providing a theoretical and practical framework for resource-efficient fruit production.

1. Introduction

Apple is one of the most economically important fruit crops worldwide, and the Loess Plateau represents a key production region in China, with average yields reaching 24.64 t ha⁻¹ in Luochuan and 18.42 t ha⁻¹ in Mizhi (Wen et al., 2022). However, this region contends with dual stressors: temporally uneven rainfall (75% concentrated in July–August) causing transient drought during fruitlet development and reduced precipitation use efficiency (30.2% lower in 2018 than 2019), alongside progressive soil desiccation forming irreversible dry layers by year 13 in Luochuan and year 7 in Mizhi. The resultant drying rates (64.6 and 68.03 mm·a⁻¹, respectively) indicate that optimal orchard productivity declines after ~23 years (Song et al., 2020; Wen et al., 2022; Zhang

et al., 2020; Zhao et al., 2015). These coupled water limitations create a paradox in which high initial productivity is achieved at the cost of long-term soil water sustainability, posing a major challenge to the stable development of apple production systems on the Loess Plateau. Developing economically efficient production management strategies capable of improving water utilization while mitigating soil desiccation has therefore become a central scientific and agricultural priority.

Super absorbent polymers (SAP) have been extensively investigated in agricultural contexts for their capacity to enhance soil properties, crop performance, and stress tolerance. SAPs enhance agroecosystem resilience by concurrently improving soil hydro-physical properties and plant stress responses, as evidenced by long-term application in winter wheat fields significantly increasing >0.25 mm soil aggregates, soil

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enzyme activity, and water use efficiency (Yang et al., 2022). Integrated strip cropping with reduced row spacing and SAP amendment under drip irrigation elevates oat (*Avena sativa*) yield by 16.65% and water productivity by 19.70% through optimized hydraulic management (Tian et al., 2022). Co-application with biofertilizers mitigates drought effects in wheat and cucumber by improving germination (Li et al., 2019), while starch-based SAP increases tree survival rates by 40% in arid areas (Lertsarawut et al., 2021). Soil moisture conservation using SAPs improves vitamin C and soluble sugar content in purple garlic (*Allium sativum*) (Pan et al., 2024), and radiation-modified Na-CMC/-PAAm hydrogels precisely regulate swelling behavior to enhance drought tolerance in sugar beet (*Beta vulgaris*) (El-Diehy et al., 2025).

Microbial fertilizers (MF) are bio-based agricultural amendments whose effectiveness is primarily mediated by metabolically active microorganisms rather than direct nutrient inputs. These inoculants promote plant performance through key functional processes, including biological nitrogen fixation, phosphate solubilization, production of plant growth-promoting compounds (e.g., phytohormones), and suppression of soil-borne pathogens, thereby regulating soil-plant interactions and nutrient cycling. MF formulations typically comprise fungal and bacterial agents. Fungal inoculants, such as *Trichoderma* spp., can alleviate apple replant disease while stimulating plant growth, whereas arbuscular mycorrhizal fungi enhance plant tolerance to abiotic stresses by improving nutrient and water acquisition (Wang et al., 2022). Bacterial agents, particularly *Bacillus* (Chen et al., 2022; Yang et al., 2022), *Pseudomonas* (Gao et al., 2025), *Streptomyces* (Abbasi et al., 2022; Santos-Medellín et al., 2021), and *Rhizobium* species (Gen-Jiménez et al., 2025), have been widely reported to enhance crop productivity through pathogen suppression, improved nutrient availability, and rhizosphere functional regulation across diverse cropping systems. Collectively, microbial fertilizers exert multifaceted phyto-beneficial effects, including increased biomass production and yield, enhanced nitrogen, phosphorus, and potassium transformation efficiency (Baas et al., 2016; Kaur et al., 2023), suppression of diseases such as *Fusarium* wilt, and improved tolerance to environmental stresses such as drought and salinity (Chandrasekaran et al., 2021; Dong et al., 2024). These functions highlight the potential of MF as a sustainable strategy for improving crop productivity under resource-limited and stress-prone agroecosystems.

Despite well-documented individual benefits of SAP in improving soil water retention and MF in enhancing rhizosphere functioning, whether their combined application generates coordinated regulation within soil-plant-microbe systems of perennial orchards remains largely unresolved. Most existing studies have been conducted using herbaceous crops, limiting mechanistic understanding in woody fruit trees, where progressive soil desiccation and chronic water limitation fundamentally reshape root architecture and rhizosphere ecological processes (Chen et al., 2021; Kaur et al., 2023). In this study, we investigated the co-application of SAP and MF in apple orchards on the Loess Plateau. We hypothesized that integrating soil water regulation with microbial functional enhancement would improve soil physicochemical conditions and rhizosphere functioning, thereby promoting plant growth, fruit quality, and productivity under rain-fed field conditions. Specifically, this study aimed to (i) assess the responses of apple growth and yield formation to SAP-MF co-application, (ii) elucidate the soil-microbe-root interactions underlying SAP-MF effects, and (iii) evaluate associated changes in soil physicochemical properties and rhizosphere microbial communities to support sustainable orchard management in water-limited agroecosystems.

2. Materials and methods

2.1. Experimental site and weather conditions

The experiment was conducted at the Zizhou Station of the Yulin Mountain Apple Experimental Station, Northwest A&F University. The

station is located in the Qingshuigou modern agricultural production area of Zizhou County, Shaanxi Province, China, at elevations ranging from 863 to 1045 m above mean sea level (37°30' N, 109°54' E, Fig. 1A and B). The soil is classified as Huangmian soil (loessial soil), texturally a silt loam. The site has a temperate continental monsoon climate with abundant sunshine and large diurnal temperature variation, a mean annual temperature of 9.2°C, a mean annual rainfall of 427.5 mm, and a frost-free period of 170 d. Meteorological monitoring at the experimental orchard showed that the effective rainfall over the whole apple growth period was 582.9 mm in 2022; in 2023, the effective rainfall over the whole growth period was 436.5 mm (Table 1). The site represents a typical rain-fed apple production system in the semi-arid Loess Plateau region.

2.2. SAP and MF treatments

The field experiment was conducted during the 2022 and 2023 growing seasons at the Zizhou Station described above. The orchard was established under a dwarf high-density planting system with wide row spacing (2.0 m × 4.0 m). Eight-year-old apple trees (*Malus domestica*) grafted as 'Hanfu'/M26/'Fupingqiu' were selected from a uniformly managed orchard. A two-factor factorial experiment was designed to evaluate the effects of a polyacrylamide-based super absorbent polymer (SAP; HRD, Gansu, China) and a microbial fertilizer (MF; Junde, Shandong, China). The SAP is a composite of polyacrylamide (PAM) and attapulgite (palygorskite), with a volumetric swelling capacity of 200–300 times its dry volume and a water-retention suction of 13–14 kg cm⁻². The product is fully biodegradable in soil within 6–7 years. Detailed specifications such as molecular weight, cross-linking density, particle size distribution, and residual acrylamide monomer content were not disclosed by the manufacturer. The MF is a commercial multi-strain compound microbial agent (fruit-tree formulation; registration No. Wei-Sheng-Wu-Fei [2012] Zhun-Zi No. 1592) comprising *Bacillus subtilis*, *Bacillus licheniformis*, *Paenibacillus mucilaginosus*, and *Trichoderma harzianum*, with a total viable count $\geq 2.0 \times 10^9$ CFU g⁻¹ and organic matter content $\geq 60\%$. The formulation also contains chitin, humic acid, fulvic acid potassium, amino acids, and trace elements (Si, Mg, Zn, B, Cu, and Fe). SAP was applied at three rates (60, 120, and 180 g plant⁻¹), and MF at three rates (250, 500, and 750 g plant⁻¹). A control treatment receiving conventional fertilization without amendments was included, resulting in a total of ten treatments (Table 2). SAP application levels were designated P1–P3 (corresponding to 60, 120, and 180 g plant⁻¹) and MF application levels were designated M1–M3 (corresponding to 250, 500, and 750 g plant⁻¹); the resulting SAP × MF combinations are hereafter referred to as PM treatments (e.g., P3M3). Each treatment was replicated three times using trees of comparable size and vigor. Amendments were applied through four holes (20 cm diameter × 60 cm depth) evenly distributed along a circular band with a radius of 1 m around each tree and thoroughly mixed with the surrounding soil (Fig. 1C and D). The same experimental trees were used throughout the two-year study. SAP and MF amendments were re-applied at the beginning of each growing season (2022 and 2023) following the same protocol. Orchard water supply relied exclusively on natural precipitation throughout the experimental period.

2.3. Soil physicochemical analysis

Soil samples were randomly collected from five points around each apple tree in each treatment using a soil sampler. Equal weights of soil were taken at 40 cm, 60 cm, and 80 cm depths, mixed thoroughly, air-dried, and sieved for subsequent analysis. Three replicate soil samples were prepared for each treatment. Soil gravimetric water content was determined by the oven-drying method. Soil pH was determined using a pH meter (Model PHS-3E pH Meter, LEI-CI, China) at a soil: distilled water ratio of 1:2.5. Soil soluble salts were determined by the residue drying method. Alkali-hydrolyzable nitrogen (AN) was determined

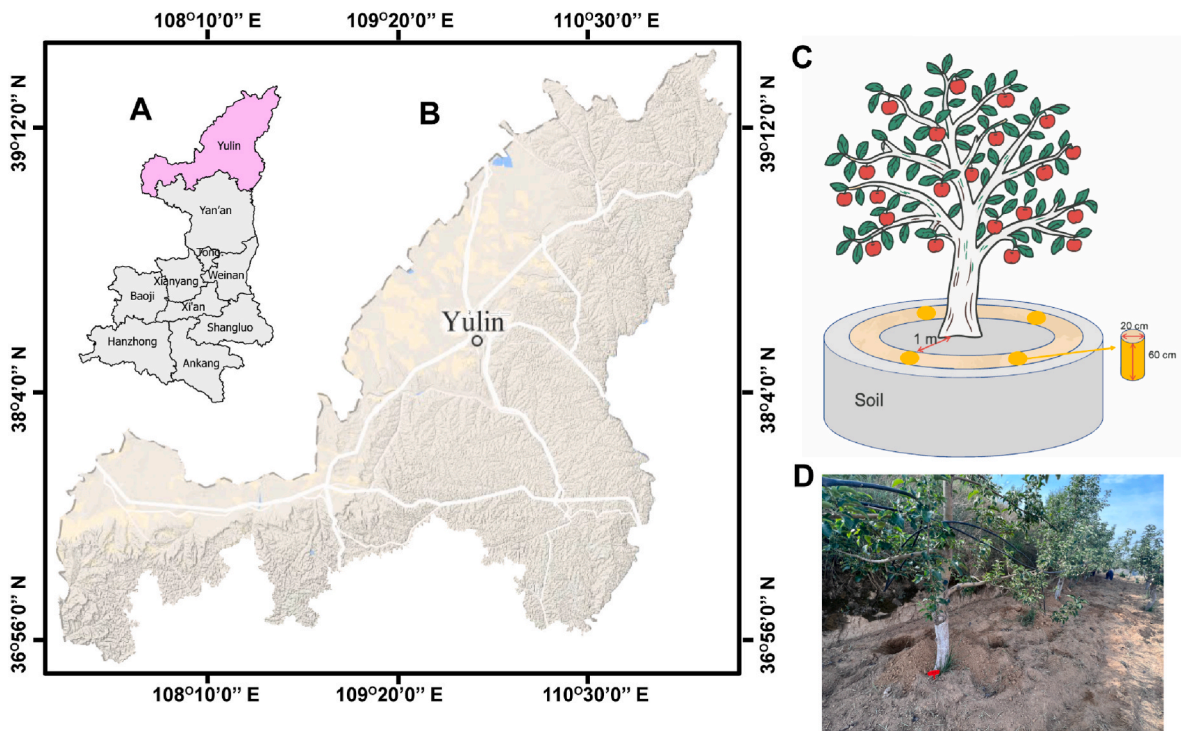


Fig. 1. Experimental setup and overview of the study area. (A) Location of the study area in northern Shaanxi Province, China (pink shaded region). (B) Location of the experimental apple orchard within Yulin City. (C) Schematic of super-absorbent polymer (SAP) and microbial fertilizer (MF) application. Materials were applied in a circular zone (radius 1 m) at a depth of 60 cm around the tree. (D) Field application of SAP and MF, showing their placement in the soil surrounding the tree base.

Table 1
Statistics of effective rainfall during the apple growth period at Zizhou Station from 2022 to 2023.

	Growth stage	Time	Effective rainfall/mm
2022	Young fruit stage	May	47.2
	Fruit expansion stage	June to August	494.5
	Fructescence	September to October	52.4
	Whole growth period	May to October	582.9
2023	Young fruit stage	May	165.6
	Fruit expansion stage	June to August	153
	Fructescence	September to October	117.9
	Whole growth period	May to October	436.5

using the alkaline diffusion method (Carter and Gregorich, 2007). Available potassium (AK) was extracted with 1 M ammonium acetate (NH₄OAc, pH 7.0) and measured by flame photometry (Han et al., 2021). Available phosphorus (AP) was extracted with 0.5 mol L⁻¹ NaHCO₃ and quantified via the molybdenum blue method (Sun et al., 2015). Soil organic content (SOC) was determined by potassium dichromate oxidation. Soil volumetric water content at the surface (12 cm probe) was measured using a Time Domain Reflectometry (TDR) system. Ten random points around apple trees per treatment were sampled.

Table 2
Dosage and application ratios of super-absorbent polymer (SAP) and microbial fertilizer (MF).

Treatment number	Control	P ₁ M ₁	P ₁ M ₂	P ₁ M ₃	P ₂ M ₁	P ₂ M ₂	P ₂ M ₃	P ₃ M ₁	P ₃ M ₂	P ₃ M ₃
PAM dosage (g/per plant)	0	60	60	60	120	120	120	180	180	180
microbial fertilizer dosage (g/per plant)	0	250	500	750	250	500	750	250	500	750

P₁, P₂, P₃: polyacrylamide-based SAP applied at 60, 120, and 180 g, respectively. M₁, M₂, M₃: MF applied at 250, 500, and 750 g, respectively. Control received no SAP or MF application.

2.4. Determination of water consumption and water use efficiency (WUE)

Water consumption (ET, mm) was determined using the soil water balance method (Sun et al., 2012):

$$ET = I + P + U - R - F \pm \Delta W \quad (1)$$

Where *I* is irrigation amount (mm), *P* is effective precipitation (mm; total rainfall minus surface runoff loss), *U* is groundwater recharge (mm), *R* is surface runoff (mm), *F* is deep percolation (mm). ΔW is the change in soil water storage (mm). Soil water storage (*W*, mm) was calculated as:

$$W = 10 \times H \times \gamma \times \theta_0 \quad (2)$$

Where *H* is soil layer thickness (cm), γ is soil bulk density (1.7 g cm⁻³), and θ_0 is gravimetric soil water content (%). Given the flat terrain, groundwater depth >30 m, and precipitation infiltration depth <2 m, *R*, *U*, and *F* were negligible and set to zero. Calculations used an *H* of 80 cm. Water use efficiency (WUE, kg·hm⁻²·mm⁻¹) was calculated as:

$$WUE = GY / ET \quad (3)$$

Where *GY* is apple fruit yield (kg·hm⁻²).

2.5. Measurement of leaf biomass, photosynthetic parameters, and SPAD value

Mature leaves were carefully sampled from three representative trees

for each treatment. From each tree, 10 leaves were randomly selected from each branch to ensure a comprehensive representation of the foliage. Five randomly chosen leaves were precisely stacked together. The total thickness of this stack was measured using a vernier caliper with a precision of ± 0.01 mm. The average thickness of a single leaf was then calculated by dividing the total stack thickness by 5. Subsequently, the average single-leaf thickness was multiplied by 100 to obtain the value equivalent to the thickness of 100 leaves. Leaf fresh weight (FW) was determined immediately after sampling using an analytical balance (± 0.01 g). For each treatment replicate, five randomly selected leaves were pooled to measure fresh weight per 5 leaves (g). Average single-leaf weight was derived from this value and scaled to express mean thickness per 100 leaves (mm).

During shoot growth cessation, 5 mature leaves per branch (three trees per treatment) were measured between 07:00 and 11:00 a.m. on sunny days using a Li-6400XT photosynthesis system (LI-COR, USA) for net photosynthetic rate (Pn), transpiration rate (E), intercellular CO₂ concentration (Ci), and stomatal conductance (Gs). Ten mature leaves per branch (three trees per treatment) were measured for SPAD values using a SPAD-502 Plus chlorophyll meter (Konica Minolta, Japan).

2.6. Determination of fruit traits

At the fruit maturity stage, 10 fruits were randomly sampled from three trees per treatment to measure fruit-related traits and all fruits from each tree were harvested to calculate yield per tree. For fruit shape index, transverse and longitudinal diameters of fruits were measured and recorded using a vernier caliper to calculate the fruit shape index. Fruit weight was measured by an electronic balance (± 0.01 g).

Fruit surface color (L*, a*, b*) was measured with a Konica Minolta CR-100 chromameter. Fifteen fruits per treatment were sampled, with five equatorial measurements per fruit: L* (lightness), a* (redness/greenness, positive values indicate redness), and b* (yellowness/blueness, positive values indicate yellowness).

Following fruit peeling, flesh samples were carefully excised from two opposing sides of the equatorial plane. The samples were homogenized using a high-speed juicer. Soluble solids content (SSC, Brix) and titratable acidity (TA, %) were then measured directly from the juice using a handheld digital refractometer-acidimeter (PAL-BX|ACID F5, Atago, Japan). For TA determination, the juice underwent a 50-fold dilution before analysis. The content of vitamin C was determined using a plant vitamin C enzyme-linked immunosorbent assay (ELISA) kit (Meimian, Jiangsu). The peel was ground thoroughly in liquid nitrogen, and 0.1 g of the sample was weighed and added to the extraction buffer. The mixture was extracted overnight at 4°C in a refrigerator, then centrifuged at low temperature, and the supernatant was collected. The absorbance values of the samples were measured at 530, 620, and 650 nm using a microplate reader, and the anthocyanin content was calculated according to the following formula: $OD (A_{530}-A_{620}) \cdot 0.1 \times (A_{650}-A_{620})$.

2.7. Fruit quality evaluation using satisfaction index

The Satisfaction Index (SI) serves as a comprehensive metric for evaluating fruit quality, representing the extent to which a fruit's expressed quality characteristics meet reasonable consumer expectations. SI is defined as a real number ranging from 0 to 1. For each apple quality indicator, the maximum observed value across all samples was assigned an SI of 1, and the minimum observed value was assigned an SI of 0. To explore the satisfaction of quality indicators of apples under different PM treatments, in this study, the average value of each indicator was used as a variable, and the relative satisfaction (M) was calculated according to formula (4).

$$M = (b_i - \text{Min}b_i) / (\text{Max}b_i - \text{Min}b_i) \quad (4)$$

Where: *M* is the Normalized Satisfaction Index for a specific quality

indicator under a PM treatment, *b_i* is the mean value of the quality indicator for the PM treatment, *Maxb_i* is the maximum mean value of the indicator observed across all studied PM treatments, *Minb_i* is the minimum mean value of the indicator observed across all studied PM treatments (Fu et al., 2023). Subsequently, a Comprehensive Satisfaction Index (*V*) was calculated by integrating the NSI values of all 11 quality indicators using a weighted summation approach:

$$V = \sum W_i M_i \quad (5)$$

Where: *V* is the comprehensive satisfaction of apple quality under different PM treatments; *M_i* is the relative satisfaction corresponding to each single index; *W_i* is the weight of the single evaluation index, and the weight also takes a value between 0 and 1. In this study, through the collection and integration of relevant literature reports on fruit quality indicators of fruit trees at home and abroad in recent years, and combined with production practice, considering the principle of emphasizing both yield and quality, the weight values of various apple indicators were determined as follows: 0.10 for Vc content in pulp, 0.10 for single - fruit weight, 0.10 for yield per plant, 0.10 for water use efficiency, 0.05 for fruit shape index, 0.15 for fruit coloring (0.05 for each of L*, a*, b*), 0.10 for soluble solids content, 0.10 for titratable acid content, and 0.10 for pulp hardness (Zhang et al., 2020; Zhao et al., 2003). The comprehensive satisfaction of various apple indicators under different PM treatments was calculated according to the formula.

2.8. RNA-seq analysis and RT-qPCR analysis

Root samples were collected from three biological replicates per treatment group, with each replicate consisting of pooled root tips (0–5 cm) from 3 individual plants. Tissues were immediately rinsed with sterile PBS to remove soil debris, blotted dry, and flash-frozen in liquid nitrogen before storage at -80°C . Total RNA was sequenced using the Illumina HiSeq platform by Novogene (Beijing, China). Then the reads were aligned to the GDDH13 reference genome version 1.1, followed by RNA-seq analysis as previously described (Shen et al., 2025). KEGG (Kyoto Encyclopedia of Genes and Genomes) analysis was performed by KOBAS software (Zhang et al., 2023). RT-qPCR analysis was carried out as described previously (Niu et al., 2022). For detecting the expression of candidate genes, a specific reverse primer based on the nucleotide difference and a common forward primer were used for RT-qPCR. Three biological replicates with three technical repeats were used. The primers used for RT-qPCR are listed in Table S1.

2.9. 16S rRNA amplicon sequencing and bacterial community analysis

Soil samples were collected from five locations and recorded as five biological replicates. The soil attached to the roots was regarded as rhizosphere soil, and the soil around the roots was regarded as conventional soil, and all were stored at -80°C . Total genomic DNA was extracted using the CTAB method, and the V3-V4 region of the bacterial 16S rRNA gene was amplified. Amplicons were sequenced on the Illumina NovaSeq platform (2 × 250 bp paired-end reads). Raw sequences were quality-filtered, denoised, and merged into Amplicon Sequence Variants (ASVs) using DADA2 (within the QIIME2 pipeline), followed by removal of low-frequency ASVs and non-bacterial taxa. Alpha diversity was evaluated using the Chao1 richness and Shannon diversity indices, with statistical significance between the Control and Treatment groups determined via the Wilcoxon rank-sum test. To assess beta diversity, principal coordinates analysis (PCoA) was performed based on Bray-Curtis dissimilarities, and community-level separation was validated through permutational multivariate analysis of variance (PERMANOVA; Adonis function in the R 'vegan' package, 999 permutations). Inter-group comparisons of bacterial taxonomic abundances at the phylum and genus levels were conducted using the Wilcoxon rank-sum test. Linear discriminant analysis effect size (LEfSe) was performed to

identify differentially abundant bacterial phyla and genera between groups. Furthermore, microbial co-occurrence networks were generated from pairwise Spearman correlations among ASVs ($|r| \geq 0.7$, Benjamini-Hochberg adjusted $P < 0.001$) to calculate key topological features (nod/edge numbers and modularity) via the 'Hmisc' and 'igraph' packages in R. Functional potentials of the bacterial communities were predicted using FAPROTAX. Statistical comparisons of the relative proportions of functional pathways between the Control and Treat groups were performed using the Mann-Whitney U test, with the 95% confidence intervals (CIs) calculated to evaluate the effect size of the differences.

2.10. Measurement of amino acids

To determine the content of free amino acids, the following procedure reported before was performed with some modifications (Silva et al., 2004). Briefly, 0.1g of ground root sample was resuspended in 50% ethanol containing 0.1 M HCl, vortex-mixed, and subjected to ultrasonic extraction at 40 Hz for 20 min. After centrifugation at 4°C and 12,000 g for 20 min, the supernatant was collected, passed through a 0.22 μm membrane filter, and analyzed via liquid chromatography-quadrupole ion trap-mass spectrometry (LC: AC, ExionLC; MS: Qtrap 5500, AB Sciex Pte. Ltd.).

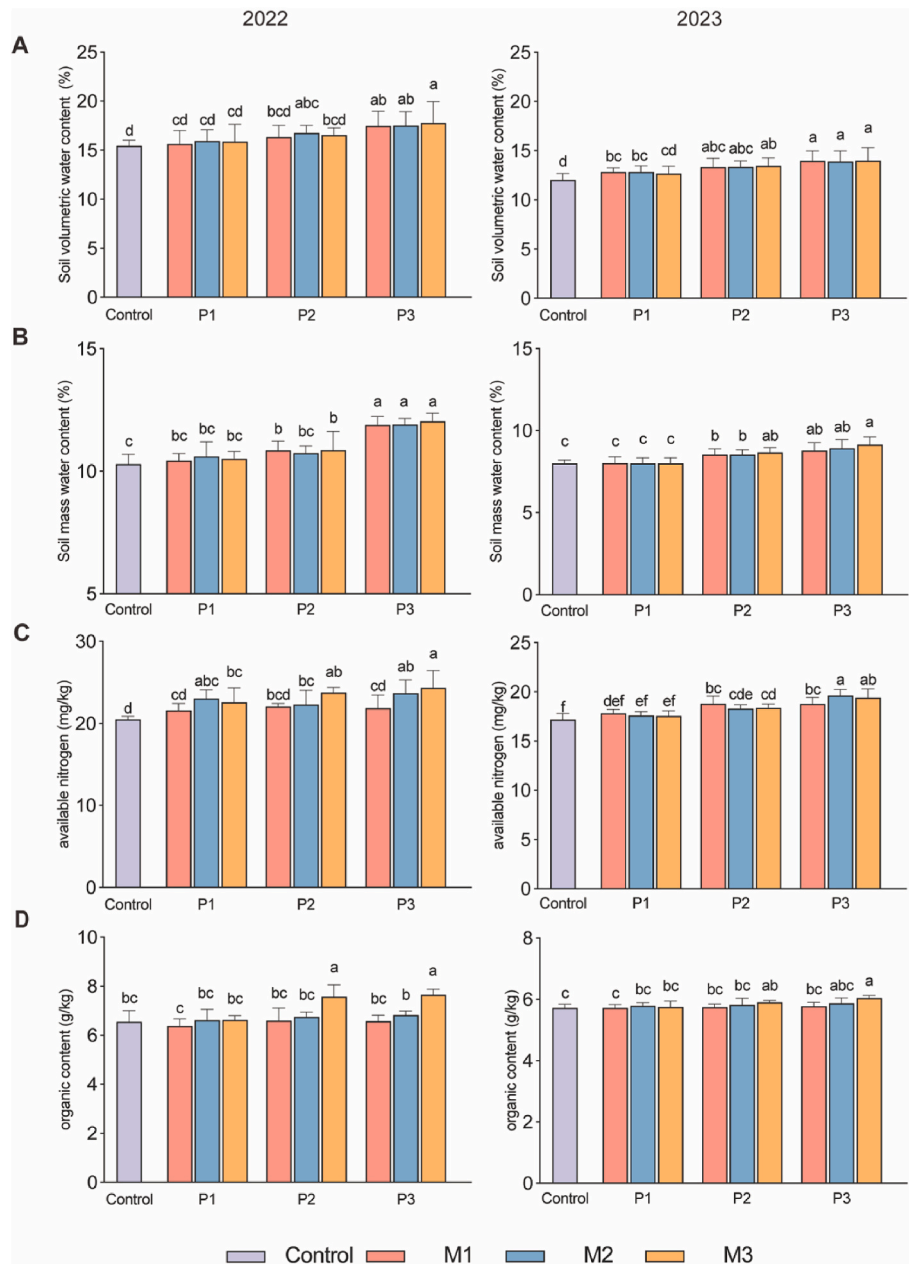


Fig. 2. Soil properties under different super absorbent polymer (P) and microbial fertilizer (M) treatments in 2022 and 2023. (A) Soil volumetric water content (%); (B) Soil mass water content (%); (C) Available nitrogen ($\text{mg}\cdot\text{kg}^{-1}$); (D) Organic content ($\text{g}\cdot\text{kg}^{-1}$). P1-P3 represent different concentrations of super absorbent polymer, and M1-M3 represent different concentrations of microbial fertilizer. Data on the left of each panel are from 2022, and data on the right are from 2023. All treatments were performed with three biological replicates. Different lowercase letters above the bars indicate significant differences among treatments, as determined by one-way ANOVA followed by Tukey's HSD test ($P < 0.05$).

2.11. Statistical analysis

Data were reported as the mean $\pm$ SD. Statistical significance was determined by Student's *t*-test or one-way ANOVA (Tukey's test) analysis using SPSS (version 22.0, IBM, USA). In addition, two-way ANOVA was performed on the nine SAP $\times$ MF factorial combinations (P1–P3 $\times$ M1–M3) to partition the main effects of SAP and MF and their interaction (SAP $\times$ MF). The Control treatment was excluded from this analysis because it falls outside the factorial design. Results of the two-way ANOVA are presented in Table S2. Differences were considered significant if $P < 0.05$, 0.01, or 0.001.

3. Results

3.1. Co-application of SAP and MF improves soil properties

To evaluate the impact of SAP and MF co-application on soil properties, field experiments were conducted in 2022 and 2023. Regarding soil moisture, both soil volumetric water content (VWC) and mass water content (SWC) exhibited a consistent upward trend with increasing concentrations of SAP (P1 to P3). In both years, the P3 treatments maintained the highest water content, significantly exceeding the control ($p < 0.05$). Notably, the P3M3 treatment achieved the peak values for both VWC and SWC. While the patterns were similar across both years, the overall soil moisture levels in 2022 were slightly higher than those recorded in 2023 (Fig. 2A and B).

The concentration of available nitrogen (AN) was significantly enhanced by the combined application of SAP and MF compared to the control (Fig. 2C). In 2022, AN levels generally increased with the intensification of both SAP and MF treatments, with P3M3 showing the highest concentration. In 2023, although the absolute AN values were lower than in 2022, a similar trend effect was observed, where P3 treatments consistently outperformed P1 and P2 across different microbial fertilizer levels (Fig. 2C). Soil organic content (SOC) also showed improvements under the experimental treatments (Fig. 2D). In 2022, significant increases in SOC were primarily observed in the higher-level treatments (P2M3 and P3M3) compared to the control. By 2023, the differences between treatments became more stabilized, with the P2M3 and P3M3 group maintaining a statistically significant advantage over the control. Overall, the combined application of high-concentration SAP and MF (P3M3) proved to be the most effective strategy for enhancing soil organic matter.

In contrast to the pronounced improvements in soil moisture, available nitrogen, and organic content, the remaining soil chemical properties were largely unaffected by the co-application of SAP and MF, and their response patterns were highly consistent between the two growing seasons (Fig. S1). Rapidly available potassium showed no significant differences among any of the treatment combinations in either 2022 or 2023, with all treatments falling into the same statistical grouping as the control (Fig. S1A). Available phosphorus likewise remained statistically uniform across treatments, exhibiting only minor, non-systematic fluctuations around the control level in both years (Fig. S1B), while soil pH was tightly buffered within a narrow alkaline range and showed no consistent treatment-dependent trend (Fig. S1C). The only discernible treatment effect among these parameters was a slight but significant reduction in soluble total salt under high SAP application, this reduction was most pronounced in the P3 treatments, where P3M3 reached the lowest salt levels in both years (Fig. S1D).

Two-way ANOVA further confirmed that SAP was the dominant factor governing soil moisture (VWC and SWC, $P < 0.0001$ in both years), whereas MF was the primary driver of AN ($P = 0.0066$ in 2022). Both factors significantly affected SOC, with a significant SAP $\times$ MF interaction detected for SOC in 2022 ($P = 0.0215$; Table S2).

3.2. SAP and MF co-application promotes leaf growth and chlorophyll content

The application of SAP and MF treatments also influenced leaf weight, thickness, and SPAD values, although the degree of statistical significance varied across parameters and years (Fig. 3). In 2022, most SAP and MF treatments led to a numerical increase in hundred-leaf weight compared to the control. Specifically, the P3M2 treatment achieved the highest weight, which was significantly greater than the control; however, other treatments such as P2M2, P2M3 and P3M3 groups showed no significant difference compared to the P3M2. In 2023, a similar trend was observed, where only the P2M2, P2M3, P3M2 and P3M3 treatments significantly outperformed the control, while other combinations remained statistically comparable to the control (Fig. 3A).

The leaf thickness showed relative stability across all treatments (Fig. 3B). In 2022, only the P3M2 and P3M3 treatments exhibited a significant increase in leaf thickness compared to the P2M2 treatment; notably, no treatments showed a statistically significant difference when compared directly to the control. By 2023, the differences among all groups further diminished, indicating no significant differences in hundred-leaf thickness across the entire experiment.

SPAD values, reflecting leaf chlorophyll content, were significantly affected by higher concentrations of SAP (Fig. 3C). In 2022, the P2 and P3 treatments (across all MF levels) significantly increased SPAD values compared to both the control and the P1 groups. However, there were no significant differences among the various M levels within the P2 and P3 groups. In 2023, the positive effect of SAP became more pronounced with increasing dosage; the P2 and P3 treatments remained significantly higher than the control.

Two-way ANOVA showed that SAP significantly affected hundred-leaf weight and SPAD values in both years, while MF had no significant main effect on any leaf parameter. No SAP $\times$ MF interaction was detected (Table S2).

3.3. SAP and MF co-application enhances the photosynthetic capacity of plants

The application of SAP and MF generally enhanced the photosynthetic capacity of apple leaves, as evidenced by increased Net photosynthesis rate (Pn), Stomatal conductance (Gs), and Transpiration rate (Tr), alongside a concomitant reduction in Inter-cellular CO₂ concentration (Ci) (Fig. 4). In 2022, Pn values were significantly enhanced by the majority of SAP and MF combinations. Specifically, all P3 treatments, along with P1M2, P1M3, P2M2, and P2M3, exhibited significantly higher Pn than the control. In contrast, the treatments with lower concentrations of both components (P1M1 and P2M1) showed no statistical difference compared to the control (Fig. 4A). In 2023, the stimulatory effect on Pn was primarily observed in the higher concentration groups. Compared to the control, all P2 and P3 treatments resulted in significantly higher Pn levels. However, no statistically significant differences were found between the P1 treatments and the control group (Fig. 4A).

Stomatal conductance exhibited a clear response to SAP and MF treatments, though the degree of significance varied between the two study years (Fig. 4B). In 2022, Gs was significantly enhanced by nearly all treatment combinations compared to the control. Specifically, the P3 treatments reached the highest Gs levels, while the P2 group and the P1M1/P1M3 combinations also maintained a statistically significant advantage over the control. In contrast, the P1M2 treatment showed no significant difference relative to the control group. In 2023, Gs was significantly enhanced by the vast majority of SAP and MF treatments compared to the control. Specifically, all P2 and P3 treatments, as well as the P1M2 combination, resulted in a significant increase in stomatal conductance relative to the control. Among the P3 treatments, P3M1 reached the peak Gs value, though P3M2 and P3M3 showed no significant difference from P3M1. In contrast, the P1M1 and P1M3 treatments

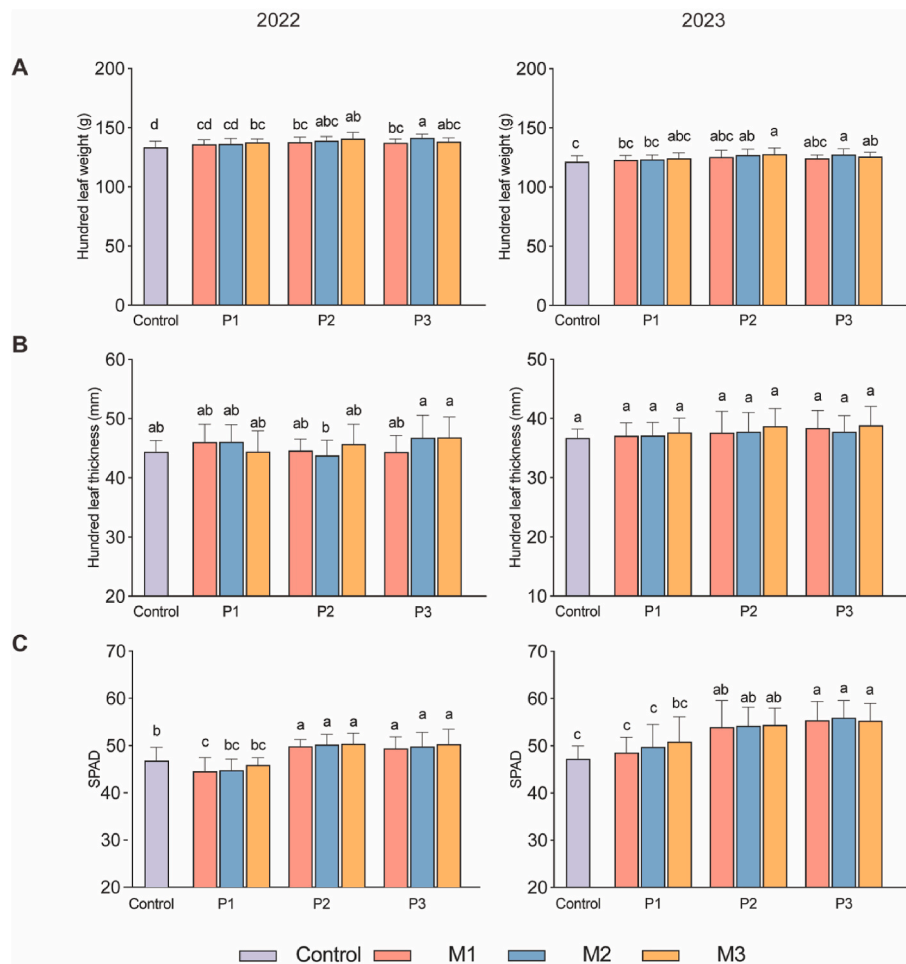


Fig. 3. Effects of super absorbent polymer (P) and microbial fertilizer (M) treatments on leaf characteristics of apple trees in 2022 and 2023. (A) Hundred - leaf weight (g), reflecting the biomass accumulation of leaves. (B) Hundred - leaf thickness (mm), showing the structural property of leaves. (C) SPAD values, indicating leaf chlorophyll content. Data on the left are from 2022, and data on the right are from 2023. All treatments were performed with three biological replicates. Different lowercase letters above the bars indicate significant differences among treatments, as determined by one-way ANOVA followed by Tukey's HSD test ($P < 0.05$).

remained statistically comparable to the control, showing no significant improvement.

Transpiration Rate (Tr) was significantly enhanced by the combined application of SAP and MF (Fig. 4C). In 2022, Tr exhibited a progressive increase with higher concentrations of SAP and MF. Compared to the control, all P2 and P3 treatments, as well as the P1M2 and P1M3 combinations, resulted in a significantly higher Tr. Specifically, the P2M3 and P3M3 treatments reached the peak values among all groups. Conversely, the P1M1 treatment showed no statistically significant difference relative to the control. In 2023, the impact of SAP was even more pronounced, with all P2 and P3 treatments showing a significant increase compared to the control group. However, no significant differences were observed between the P2 and P3 groups, regardless of the microbial fertilizer concentration applied, as they shared similar significance levels.

In contrast to the other parameters, Ci levels generally decreased with the application of SAP and MF (Fig. 4D). In 2022, P2M1, P2M2, and all P3 treatments significantly reduced Ci compared to the control. By 2023, this inhibitory effect was consistent across all experimental groups; every SAP and MF combination (P1–P3) resulted in a significantly lower Ci than the control, although no statistical differences were found among the various treatment combinations themselves.

Two-way ANOVA confirmed that SAP was the primary driver of Pn, Gs, Tr, and Ci, with no significant SAP $\times$ MF interaction for any photosynthetic parameter in either year (Table S2).

3.4. SAP and MF co-application enhances plant water use efficiency

The co-application of SAP and MF significantly influenced the water use efficiency (WUE) of yield in apple trees during both experimental years (Fig. 5). In 2022, all treatment combinations resulted in a significant increase in WUE compared to the control. The most substantial improvements were observed in the P2 treatments and the P3M2/P3M3 combinations, which achieved the highest efficiency levels among all groups. While the P1 group and the P3M1 treatment also enhanced WUE relative to the control, their effects were statistically lower than those recorded in the P2 and P3M2/P3M3 treatments. In 2023, the responses of WUE to the treatments were more varied. Compared to the control, a significant increase in WUE was observed in the P1M2, P1M3, P2M2, P2M3, and all P3 treatments, with P3M2 achieving the maximum value among all tested combinations. Conversely, the P1M1 and P2M1 treatments showed no statistically significant improvement, remaining comparable to the control group. Notably, two-way ANOVA revealed significant main effects of both SAP and MF on WUE, together with a significant SAP $\times$ MF interaction in both 2022 ($P = 0.0106$) and 2023 ($P < 0.0001$), indicating a true synergistic effect of co-application on water use efficiency (Table S2).

3.5. SAP and MF co-application improves apple orchard yield

As shown in Fig. 6, the application of different concentrations of SAP

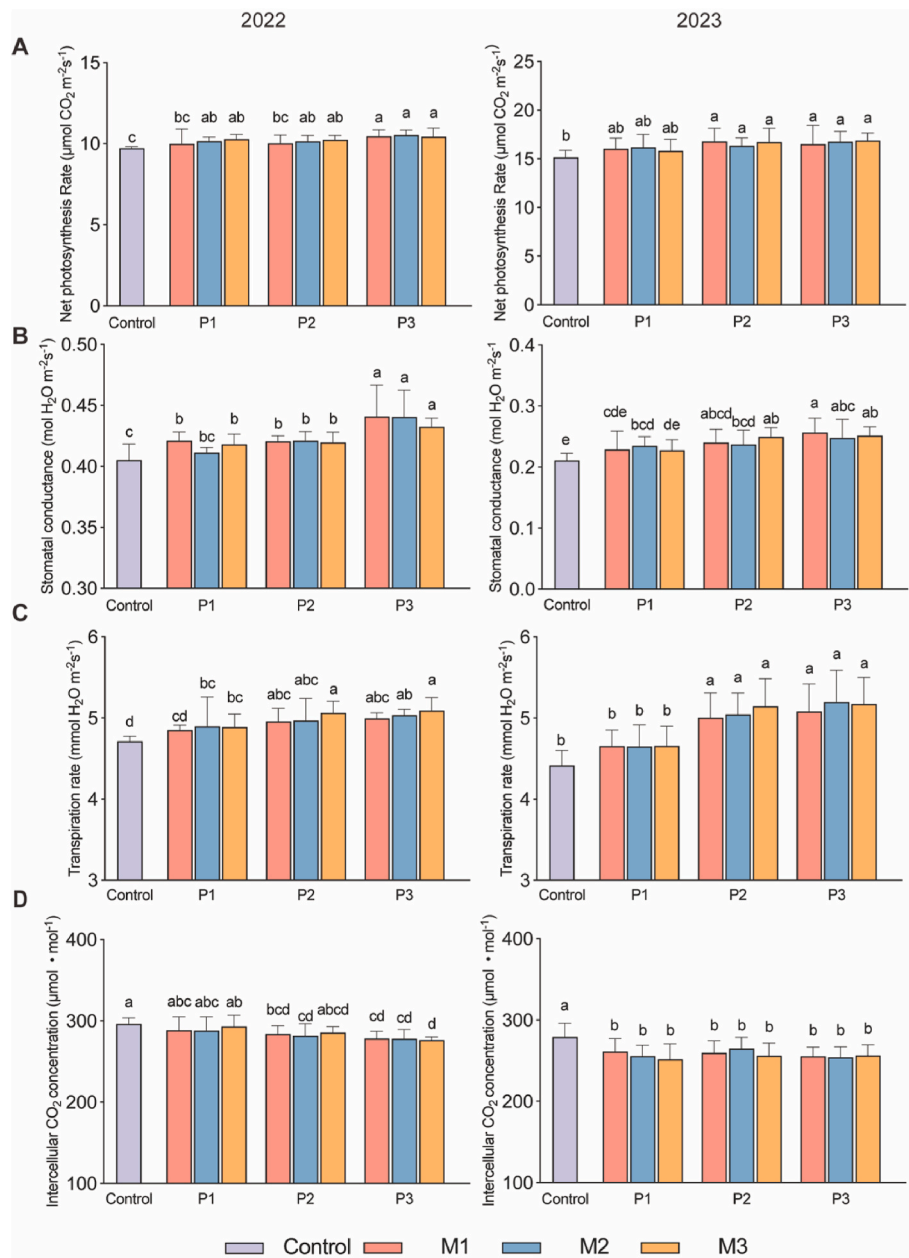


Fig. 4. Photosynthetic characteristics of apple tree leaves under different Super absorbent polymer (P) and Microbial fertilizer (M) treatments. **(A)** Net photosynthesis rate (P_n , $\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$); **(B)** Stomatal conductance (G_s , $\text{mol H}_2\text{O m}^{-2} \text{ s}^{-1}$); **(C)** Transpiration rate (T_r , $\text{mmol H}_2\text{O m}^{-2} \text{ s}^{-1}$); **(D)** Intercellular CO_2 concentration (C_i , $\mu\text{mol} \cdot \text{mol}^{-1}$). Data on the left of each panel are from 2022, and data on the right are from 2023. All treatments were performed with three biological replicates. Different lowercase letters above the bars indicate significant differences among treatments, as determined by one-way ANOVA followed by Tukey's Honestly Significant Difference (HSD) test ($P < 0.05$).

and MF positively affected the yield per tree and single fruit weight, while the quantity of fruits per tree remained largely unaffected. The treatments had a minimal impact on the quantity number per tree (Fig. 6A). In 2022, no significant differences were observed between any of the treatment groups and the control. The only significant variation occurred between the treatments with the highest fruit numbers (all P1 combinations and P2M1) and the treatment with the lowest (P3M1). In 2023, while the overall quantity of fruits increased across all groups compared to the previous year, no significant differences were detected among any of the treatments and the control. In contrast to the fruit quantity, the combined application of SAP and MF significantly increased the yield per tree in both consecutive years (Fig. 6B). In 2022, all treatment groups exhibited significantly higher yields than the control, with combinations under P2 and P3 concentrations performing

notably well. This trend continued into 2023, where the control group again recorded the lowest yield, while treatments such as P2M3 and all P3 combinations (P3M1, P3M2, P3M3) achieved the highest yield levels, significantly outperforming the control. The enhancement of single fruit weight under different treatments closely mirrored the trends observed in total yield (Fig. 6C). In 2022, treatments within the P2 and P3 groups significantly increased single fruit weight compared to the control and most P1 treatments. In 2023, the P2M3, P3M1, and P3M3 treatment groups consistently maintained the highest single fruit weights, remaining significantly superior to the control. Two-way ANOVA indicated that SAP was the dominant factor for single fruit weight ($P < 0.0001$ in 2022) and yield per tree, while MF showed no significant main effect. No SAP $\times$ MF interaction was detected for any yield parameter (Table S2).

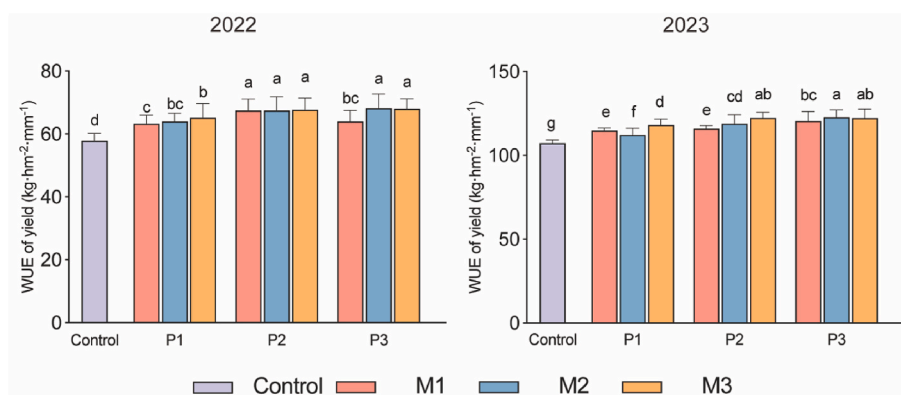


Fig. 5. Water use efficiency of yield (WUE of yield, kg·hm⁻²·mm⁻¹) of apple trees under different super absorbent polymer (P) and microbial fertilizer (M) treatments in 2022 and 2023. P1–P3 represent different concentrations of super absorbent polymer, and M1–M3 represent different concentrations of microbial fertilizer. Data on the left are from 2022, and data on the right are from 2023. All treatments were performed with three biological replicates. Different lowercase letters above the bars indicate significant differences among treatments, as determined by one-way ANOVA followed by Tukey's HSD test ($P < 0.05$).

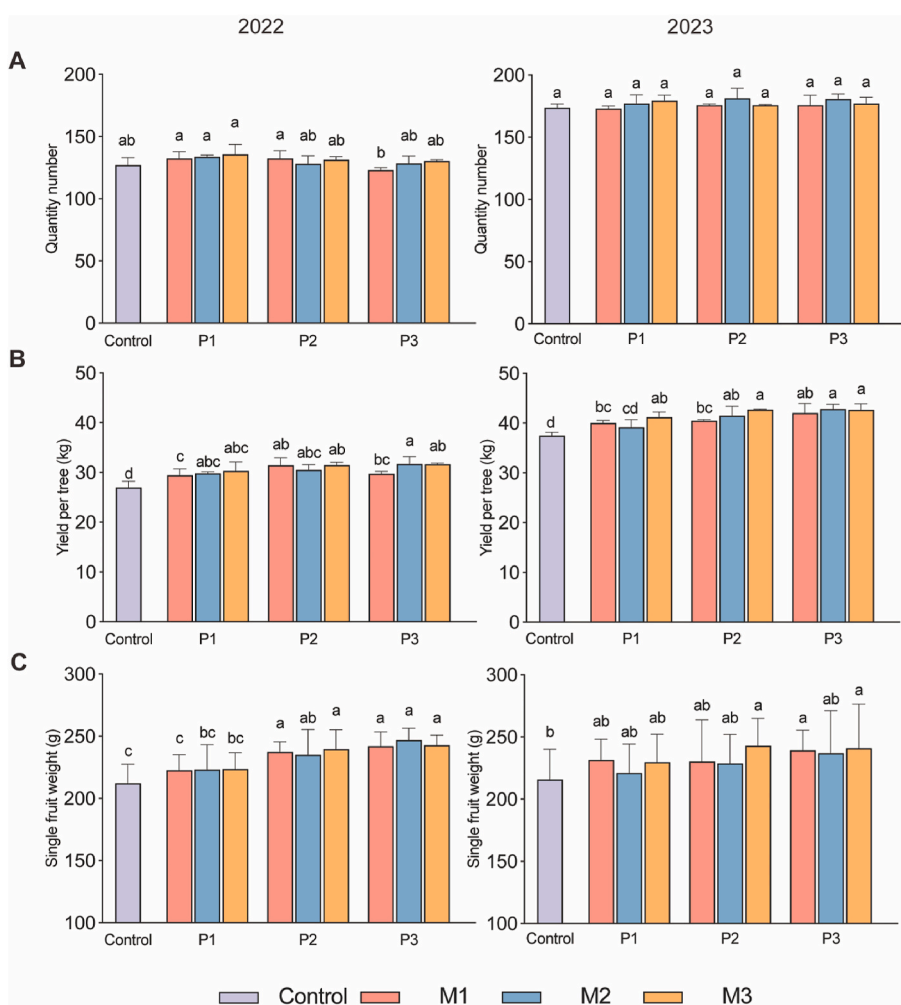


Fig. 6. Fruit-related traits of apple trees under different super absorbent polymer (P) and microbial fertilizer (M) treatments. (A) Quantity number per tree; (B) Yield per tree (kg); (C) Single fruit weight (g). P1–P3 represent different concentrations of super absorbent polymer, and M1–M3 represent different concentrations of microbial fertilizer. Data on the left of each panel are from 2022, and data on the right are from 2023. All treatments were performed with three biological replicates. Different lowercase letters above the bars indicate significant differences among treatments, as determined by one-way ANOVA followed by Tukey's HSD test ($P < 0.05$).

3.6. SAP and MF co-application alters apple fruit quality

To determine the effects of SAP and MF on fruit quality, relevant

appearance and biochemical parameters were assessed during the 2022 and 2023 seasons (Fig. S2). For pericarp coloration, treatments with high SAP concentrations (P3M1, P3M2, and P3M3) significantly

elevated the lightness parameter (L^*) compared to the control group, with the P3M3 treatment achieving the peak L^* value across both years. Furthermore, L^* exhibited an increasing tendency as the SAP application rate increased within fixed MF levels; however, variations in MF concentrations at constant SAP levels produced no significant chromatic shifts. In contrast to lightness, neither the a^* (red-green) nor the b^* (yellow-blue) color parameters differed significantly from the control group across any of the treatments (Fig. S2A). Consistent with the stable a^* (redness) values, the anthocyanin content in the pericarp remained comparable to that of the control group. Although minor fluctuations were observed among certain treatment combinations (e.g., between P2M2 and P2M3), no treatments resulted in a statistically significant difference in anthocyanin content when compared directly to the control across both seasons (Fig. S2B).

Furthermore, the treatments significantly influenced internal fruit quality parameters (Fig. S3). Soluble solids content (SSC) was notably enhanced in the second year of application, while no significant differences were observed in 2022, combinations with moderate to high concentrations (specifically P2M3, P3M2, and P3M3) significantly increased SSC compared to the control in 2023 (Fig. S3A). In contrast, titratable acidity remained remarkably stable across all treatments, with no significant differences observed between any treatment and the control in either season (Fig. S3B). The Ratio of total soluble solids to titratable acidity (RTT) parameter exhibited year-to-year variation (Fig. S3C). In 2022, most treatments showed no significant difference compared to the control, except for the P2M1 and P3M1 treatments, which resulted in a significant decrease. Conversely, in 2023, the majority of the treatments significantly enhanced RTT compared to the control, with the exception of the P2M1 and P2M2 groups, which remained statistically comparable to the Control. Finally, regarding pulp Vitamin C content, the highest concentration combination (P3M3) consistently yielded optimal results, significantly elevating Vitamin C levels compared to the control across both consecutive years, whereas other treatments remained statistically similar to the Control (Fig. S3D).

Two-way ANOVA showed that MF was the primary driver of SSC ($P < 0.0001$ in both years), whereas SAP dominated L^* and fruit dimensions. No significant SAP $\times$ MF interaction was found for any fruit quality parameter (Table S2).

3.7. Effects of SAP and MF co-application on the satisfaction index of apple fruits

The application of SAP and MF effectively promoted fruit expansion, as evidenced by significant increases in both longitudinal and equatorial diameters, particularly under higher SAP concentrations (Fig. S4A and B). In 2022, treatments with moderate to high SAP levels (most P2 combinations and all P3 combinations) significantly increased both longitudinal and equatorial diameters compared to the control group.

This growth-promoting trend remained consistent in 2023, where the highest SAP concentration group (P3M1, P3M2, and P3M3) consistently maintained significantly larger longitudinal and equatorial diameters than the control. In contrast, the low-concentration treatments (P1 group) showed limited effects on fruit expansion, generally remaining statistically comparable to the control across both seasons. Despite the significant increases in absolute fruit dimensions under high SAP treatments, the fruit shape index remained remarkably stable (Fig. S4C). Across all treatment combinations in both the 2022 and 2023 seasons, no significant differences in the shape index were observed compared to the control group. This indicates that the P and M treatments promoted a proportional and uniform expansion of the fruit without altering its intrinsic shape characteristics.

To comprehensively assess the effects of different treatments on fruit quality, yield, and water use efficiency (WUE) in 2022, a multidimensional evaluation was conducted based on the normalized indices of various parameters (Table 3). The results demonstrated that the treatments significantly influenced the overall performance. The P3M3 treatment exhibited the best comprehensive performance, achieving the highest overall satisfaction score of 0.817 and ranking first among all groups. Under this optimal treatment, key indicators including Vc (1.00), single fruit weight (1.00), yield (0.99), and WUE (0.98) reached or were proximate to their maximum values. The P2M3 and P3M2 treatments followed, ranking second and third with overall satisfaction scores of 0.757 and 0.725, respectively. Notably, P2M3 showed superior performance in SSC (1.00) and TA (1.00), while P3M2 maximized WUE (1.00) and Firmness (1.00). Conversely, the control group recorded the lowest overall satisfaction score (0.182, ranking 10th), with baseline minimums across critical metrics such as Vc, single fruit weight, yield, and WUE.

To further evaluate the effects of various treatments on fruit quality, yield, and water use efficiency (WUE) in 2023, a comprehensive multidimensional analysis was performed using normalized indices (Table 4). The results indicated substantial variations in overall performance across treatments. Specifically, the P3M3 treatment achieved the highest overall satisfaction score (0.850) and ranked first among all groups. Under this optimal treatment, several critical quality parameters, including Vc, L^* , SSC, and titratable acidity (TA), reached their maximum normalized values (1.00), while yield (0.97) and WUE (0.97) were also near optimal. Following closely, the P3M2 and P2M3 treatments ranked second and third, with overall satisfaction scores of 0.822 and 0.708, respectively. Notably, P3M2 maximized yield (1.00), WUE (1.00), and firmness (1.00), whereas P2M3 showed peak performance in single fruit weight (1.00) and fruit shape index (1.00). Conversely, the control group exhibited the poorest comprehensive performance, recording the lowest overall satisfaction score (0.078, ranking 10th) with baseline minimums across most key indicators.

Table 3
Comprehensive evaluation of fruit quality and water use efficiency under different treatments in 2022.

Treatment	Control	P ₁ M ₁	P ₁ M ₂	P ₁ M ₃	P ₂ M ₁	P ₂ M ₂	P ₂ M ₃	P ₃ M ₁	P ₃ M ₂	P ₃ M ₃
Vc	0	0.18	0.21	0.13	0.33	0.41	0.28	0.45	0.49	1.00
Single fruit weight	0	0.3	0.31	0.32	0.73	0.66	0.79	0.85	0.88	1.00
Yield	0	0.53	0.6	0.71	0.94	0.75	0.95	0.59	1.00	0.99
WUE	0	0.52	0.6	0.71	0.93	0.93	0.95	0.59	1.00	0.98
Fruit shape index	0.49	0.4	0.28	0.58	0.64	0.59	0.00	1.00	0.84	0.77
L^*	0.04	0	0.27	0.15	0.23	0.32	0.18	0.54	0.79	1
a^*	0.61	1	0.15	0.22	0.00	0.19	0.79	0.74	0.14	0.41
b^*	0.18	0.12	0.04	0.28	0.85	0.21	1	0.5	0.38	0
SSC	0.73	0.37	0.42	0.55	0	0.92	1.00	0.11	0.67	0.83
TA	0.35	0	0.46	0.37	0.24	0.48	1	0.37	0.13	0.61
Firmness	0.08	0.46	0.3	0.82	0	0.24	0.66	0.32	1	0.68
Overall Satisfaction	0.182	0.365	0.387	0.494	0.497	0.589	0.757	0.526	0.725	0.817
Ranking	10	9	8	7	6	4	2	5	3	1

Vc, vitamin C content; WUE, water use efficiency; L^* , a^* , b^* , color parameters (lightness, redness, yellowness); SSC, soluble solids content; TA, titratable acidity. P₁, P₂, P₃ represent three SAP (super absorbent polymer) application rates; M₁, M₂, M₃ represent three microbial fertilizer application rates.

Table 4

Comprehensive evaluation of fruit quality and water use efficiency under different treatments in 2023.

Treatment	Control	P ₁ M ₁	P ₁ M ₂	P ₁ M ₃	P ₂ M ₁	P ₂ M ₂	P ₂ M ₃	P ₃ M ₁	P ₃ M ₂	P ₃ M ₃
Vc	0	0.17	0.02	0.05	0.46	0.22	0.16	0.7	0.58	1
Single fruit weight	0	0.58	0.19	0.51	0.54	0.48	1	0.87	0.78	0.92
Yield	0	0.48	0.31	0.7	0.56	0.75	0.98	0.86	1	0.97
WUE	0	0.48	0.31	0.7	0.56	0.75	0.97	0.86	1	0.97
Fruit shape index	0.14	0	0.29	0.39	0.59	0.42	1	0.41	0.63	0.53
L*	0	0.05	0.03	0.02	0.29	0.4	0.35	0.8	0.77	1
a*	0	0.25	0.02	0.16	0.47	0.99	0.46	0.73	1	0.73
b*	0.67	0.76	0.6	1	0	0.56	0.38	0.18	0.79	0.34
SSC	0	0.06	0.26	0.27	0.09	0.4	0.88	0.35	0.83	1
TA	0.37	0	0.16	0.53	0.14	0.67	0.33	0.45	0.43	1
Firmness	0	0.65	0.18	0.53	0.29	0.14	0.69	0.09	1	0.37
Overall Satisfaction	0.078	0.343	0.221	0.478	0.388	0.535	0.708	0.610	0.822	0.850
Ranking	10	8	9	6	7	5	3	4	2	1

Vc, vitamin C content; WUE, water use efficiency; L*, a*, b*, color parameters (lightness, redness, yellowness); SSC, soluble solids content; TA, titratable acidity. P₁, P₂, P₃ represent three SAP (super absorbent polymer) application rates; M₁, M₂, M₃ represent three microbial fertilizer application rates.

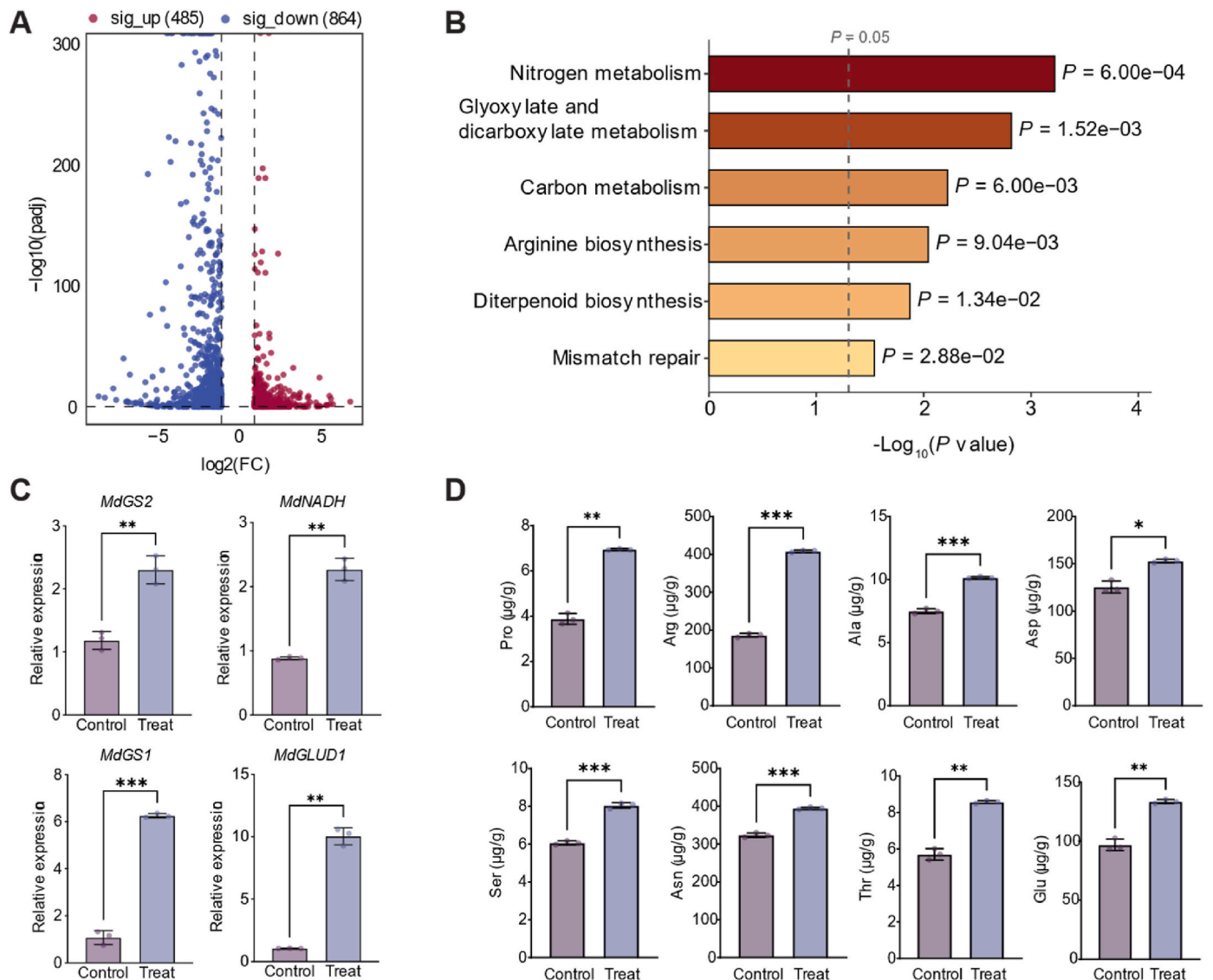


Fig. 7. Comparison of differentially expressed genes (DEGs) in P3M3 treatment and control. (A) Volcano plot of DEGs (Treatment vs Control); genes with $padj \leq 0.05$ and $|\log_2FC| \geq 1$ are significant. (B) KEGG pathway enrichment of upregulated DEGs. (C) RT-qPCR validation of four nitrogen metabolism and arginine biosynthesis-related genes. (D) Free amino acid content in roots. Data are means $\pm$ SD ($n = 3$ biological replicates). Statistical significance was determined by two-tailed Student's t-test: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

3.8. Transcriptome analysis of apple root systems under SAP and MF co-application

To further explore the molecular mechanism of apple root systems under SAP and MF co-application, we conducted transcriptome analysis between the control group and P3M3 treatment. Based on the screening criteria of an adjusted P -value (padj) < 0.05 and $|\log_2(\text{Fold Change})| > 1$, a total of 1349 differentially expressed genes (DEGs) were identified. Among these, 485 genes were significantly upregulated and 864 genes were downregulated in the P3M3 treatment compared to the control (Fig. 7A). To understand the biological functions of the upregulated DEGs, KEGG pathway enrichment analysis was performed. The results revealed that these genes were highly enriched in essential metabolic pathways, notably carbon metabolism, nitrogen metabolism, and arginine biosynthesis (Fig. 7B). Given the critical role of nitrogen assimilation, we further examined the expression patterns of key DEGs involved in nitrogen metabolism and arginine biosynthesis. Comparative analysis of transcript abundance (FPKM) demonstrated that several critical genes, including *MdGLUD1*, *MdGS1*, *MdGS2*, and *MdNADH*, were significantly upregulated in the roots under the P3M3 treatment (Fig. S5). To validate the reliability and accuracy of the transcriptomic data, quantitative real-time PCR (qRT-PCR) was performed to examine the expression levels of these four key nitrogen metabolism-related genes (Fig. 7C). Consistent with the transcriptomic upregulation of nitrogen-related genes, the accumulation of free amino acids in the roots was markedly enhanced. The concentrations of all measured amino acids (Pro, Arg, Ala, Asp, Ser, Asn, Thr, and Glu) were significantly higher in the P3M3-treated roots than in the control. Notably, Arg, Ala, Ser, and Asn exhibited the most dramatic increases ($P < 0.001$) (Fig. 7D). These findings suggest that the P3M3 treatment robustly activates nitrogen metabolism and amino acid biosynthesis at both the transcriptional and metabolic levels.

3.9. SAP and MF co-application restructures the bulk soil and rhizosphere bacterial community

Given the central role of bacteria in soil nutrient cycling and root nitrogen assimilation, we further characterized the bacterial communities of the rhizosphere and bulk soil in the P3M3 and control groups via 16S rRNA amplicon sequencing.

In the bulk soil, the P3M3 treatment did not significantly alter bacterial alpha diversity, as both Chao1 richness and the Shannon index remained comparable to the control (Fig. S6A). Nevertheless, PCoA based on Bray-Curtis dissimilarity revealed a clear separation between the TreatSoil and ControlSoil communities along PCoA1 (65.6% of variance explained), which was statistically supported by PERMANOVA (Fig. S6B). Proteobacteria and Actinobacteriota remained the dominant bacterial phyla in both groups, while Bacteroidota and Myxococcota were significantly enriched and Thermoplasmata and Patescibacteria were significantly depleted under P3M3 treatment (Fig. S6C). Among genera, *Devosia*, *Sphingomonas*, and *Streptomyces* increased significantly, whereas *MND1*, *Variovorax*, *Gaiella*, and *Ensifer* decreased significantly under treatment (Fig. S6C). LEfSe analysis further identified Proteobacteria, Bacteroidota, and Myxococcota as the phyla most enriched in the treated bulk soil, together with a broader set of differentially abundant genera with documented plant growth-promoting activity, including *Devosia*, *Sphingomonas*, *Streptomyces*, *Flavobacterium*, *Lyso-bacter*, *Sphingobacterium*, and *Brevundimonas* (Fig. S6D).

In the rhizosphere, P3M3 did not significantly affect bacterial alpha diversity, with Chao1 richness and the Shannon index both comparable to the control (Fig. 8A). Community structure was another matter. PCoA separated TreatRhi from ControlRhi cleanly, with PCoA1 alone accounting for 73.0% of the variance and PERMANOVA confirming the split (Fig. 8B). The co-occurrence network showed a comparable pattern of reorganization: TreatRhi exhibited slightly higher modularity than ControlRhi (0.9813 vs. 0.9717), suggesting a more compartmentalized

community structure under treatment (Fig. 8C). At the taxonomic level, Actinobacteriota expanded significantly in TreatRhi compared with ControlRhi, whereas Proteobacteria and Gemmatimonadota contracted (Fig. 8D). This taxonomic shift was further reflected at the genus level by the enrichment of the actinobacterial genus *Nocardioideae* and the concomitant depletion of *Novosphingobium*. LEfSe further identified the actinobacterial genera *Frankia*, *Promicromonospora* and *Pseudonocardia* as enriched under treatment (Fig. 8E). Notably, several proteobacterial genera were enriched despite the general contraction of their parent phylum, including the diazotrophic *Bradyrhizobium* and the plant growth-promoting *Pseudomonas* (Fig. 8E). FAPROTAX provided a functional layer to these taxonomic shifts, showing an enrichment in nitrogen fixation and nitrite respiration pathways alongside a concomitant decline in xylanolysis and chitinolysis (Fig. S7).

4. Discussion

Water scarcity and low soil fertility are the core bottlenecks for sustainable apple industry development in the Loess Plateau of China. SAP and MF serve as agricultural soil conditioners (Buchmann et al., 2025; Wei et al., 2024), while their combined application in apple cultivation has not been reported. In the present study, the co-application of SAP and MF improves soil properties and enhances plant growth, fruit yield, fruit quality, WUE, and also reshaping the rhizosphere microbial community.

SAP and MF are widely used soil conditioners in agricultural production (Buchmann et al., 2025; Wei et al., 2024). SAP increased soil moisture content to provide a suitable environment for microbial activity, while MF enhanced nutrient availability (available nitrogen), increased soil organic matter (SOM), and reduced total soluble salts. This integrated response generated a three-fold complementary improvement in soil water status, enhanced fertility and mitigated salinization. Specifically, the increase in SOM not only directly enhanced soil fertility but also strengthened the soil's water and nutrient retention capacity by promoting the formation of soil aggregates (Wang et al., 2019; Wu et al., 2024; Yang et al., 2022). The reduction in total soluble salts helped alleviate salt stress. In addition, SAP was the dominant factor governing soil moisture dynamics, whereas MF primarily drove SOM accumulation and nutrient activation. This complementary functional specialization collectively optimized soil properties.

Leaves, as the primary organs of photosynthesis, critically determine the accumulation of photo assimilates (Li et al., 2023). Our findings demonstrate that the combined SAP and MF treatment significantly enhanced SPAD, Pn, E, and Gs, while concurrently reducing the Ci. These results collectively indicate that the treatment effectively improved both the light reactions and carbon assimilation efficiency, thereby boosting photosynthetic output. The dosage effect of SAP was significant, with SPAD, Pn, and Gs exhibiting significant increases in response to higher SAP application rates. This enhancement is closely linked to SAP-mediated improvements in soil water availability and mitigation of drought stress (El-Diehy et al., 2025; Yang et al., 2022). Meanwhile, MF contributed primarily by activating soil nutrients, notably available nitrogen, thereby supplying essential precursors for chlorophyll biosynthesis and enhancing photosynthetic enzyme activity (Demir et al., 2024; Galindo et al., 2024). The complementary interaction between SAP and MF established an operational nexus of enhanced hydraulic regulation, optimized nutrient provisioning, and improved photosynthetic resource use efficiency (Li et al., 2019). The observed increase in fresh weight per 100 leaves further corroborates enhanced biomass accumulation in leaves. In contrast, the lack of significant alterations in leaf thickness suggests that this specific combination exerts limited effects on modulating leaf anatomical structure, with its core benefits instead focused on enhancing Pn and biomass production.

Yield and quality indicators are direct manifestations of the treatment effects (Wang et al., 2016). The combination of SAP and MF significantly increased the single fruit weight, yield per plant, and fruit

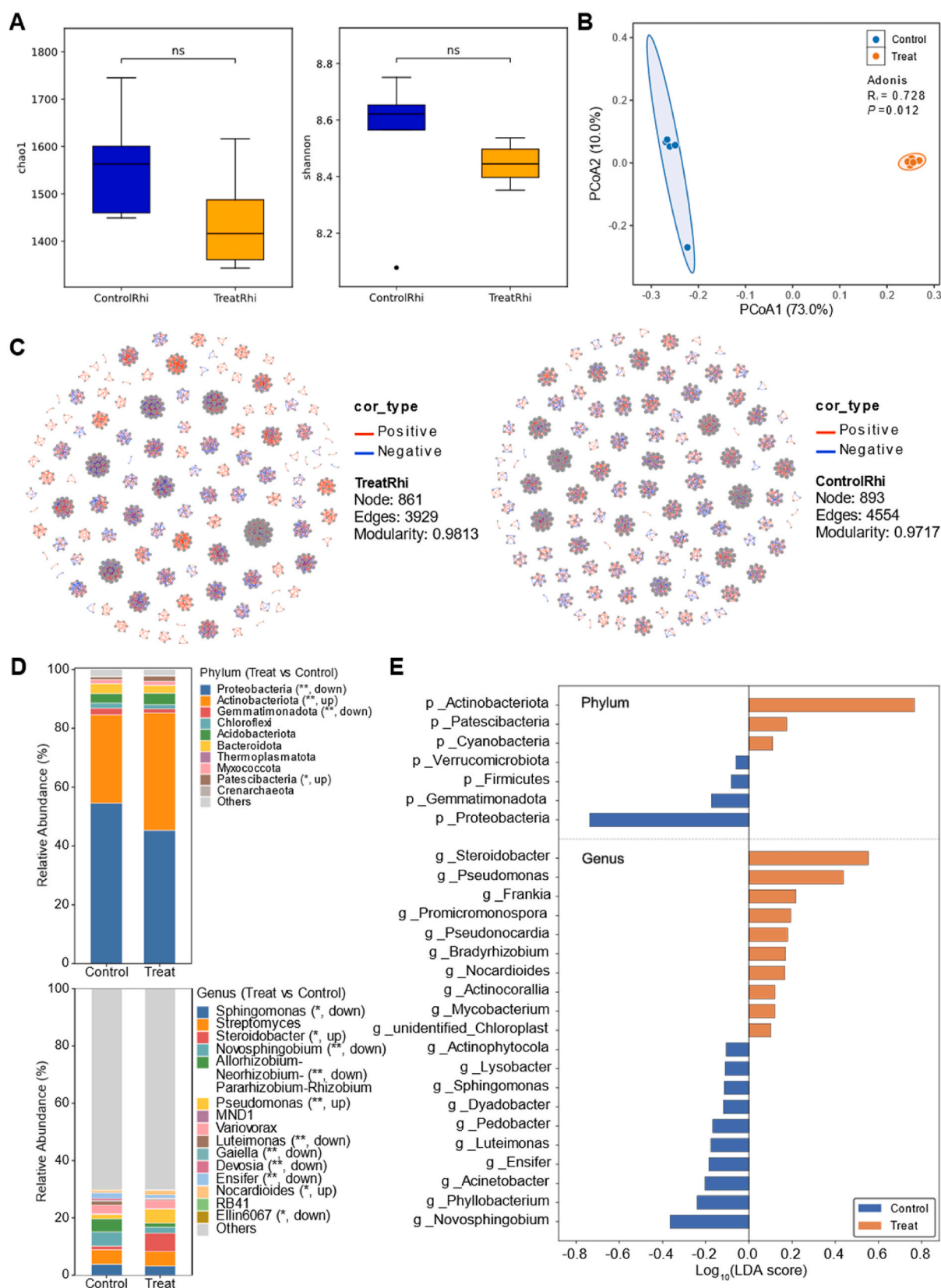


Fig. 8. Comparison of the rhizosphere bacterial community between P3M3 treatment and control. **(A)** Alpha diversity (Chao1 richness and Shannon index) of the rhizosphere bacterial community in Control and Treat groups. **(B)** Principal coordinates analysis (PCoA) based on Bray-Curtis dissimilarity, with group separation tested by PERMANOVA (Adonis). **(C)** Co-occurrence networks of the rhizosphere bacterial community in Treat and Control groups; red and blue edges indicate positive and negative correlations, respectively. **(D)** Relative abundance of dominant bacterial taxa at the phylum and genus levels in Control and Treat groups. **(E)** LEfSe analysis identifying differentially enriched bacterial phyla and genera between Control and Treat groups. Data are means $\pm$ SD ($n = 5$ biological replicates). Statistical significance was determined by the Wilcoxon rank-sum test (A, D; ns = not significant) and PERMANOVA (Adonis, B): * $P < 0.05$, ** $P < 0.01$.

transverse/longitudinal diameter. Among them, the treatments of P3M2 and P3M3 showed the best performance. This improvement is closely related to the increased Pn and available nitrogen. In terms of regulatory patterns, SAP plays a dominant role in promoting fruit enlargement (increase in transverse/longitudinal diameter). At the same MF level, fruit size showed a positive correlation with SAP. On the other hand, the Vc and SSC in the P3M3 treatment increased significantly. This is associated with the MF-mediated promotion of carbon-nitrogen metabolite accumulation in fruits (Mosa et al., 2016; Zhang et al., 2022). In addition, the increase in fruit L* may be associated with the improvement of water status by SAP and the reduction of epidermal stress damage (Ji et al., 2024). The fruit shape index and anthocyanin content were not significantly affected, indicating that this combination has selective regulation on the fruit appearance morphology.

In the arid and semi-arid regions of Northwest China, water is the core limiting factor for apple production (Ding et al., 2024). This study has confirmed that the combination of SAP and MF significantly improved WUE, with the P3M2 treatment showing the best performance. This effect is mainly attributed to the water retention capacity of SAP. SAP bolsters the water-holding capacity of soil, as evidenced by a significant increase in soil gravimetric and volumetric water content. It also curtails unproductive evaporation and facilitates the efficient uptake of water by roots (Yang et al., 2022). Conversely, MF may indirectly enhance water absorption efficiency by modifying the soil structure, for instance, through an increase in organic matter content, and by augmenting root vitality (Galindo et al., 2024). Indeed, two-way ANOVA confirmed a significant SAP $\times$ MF interaction for WUE in both years (2022, $P = 0.0106$; 2023, $P < 0.0001$; Table S2), indicating that co-application produced a genuinely synergistic improvement in water use efficiency. The synergistic interplay between SAP and MF establishes a self-reinforcing cycle of optimized water conservation and improved plant WUE, providing a viable strategy for water-saving and yield-intensified apple production in arid and semi-arid regions.

Notably, the two experimental years differed considerably in precipitation amount and seasonal distribution (Table 1): effective rainfall totaled 582.9 mm in 2022, with 84.8% concentrated in June–August, whereas 2023 received only 436.5 mm distributed more evenly across growth stages. Despite these differences, the main effects of SAP and MF remained highly consistent across years (Table S2), indicating that the complementary roles described above appear robust across contrasting rainfall regimes. The SAP $\times$ MF interaction for WUE, as noted above, was significant in both years but markedly stronger in the drier 2023 (Table S2), suggesting that the synergistic benefit on water use efficiency appears to be amplified under intensified water limitation. In addition, SSC gained a significant SAP main effect in 2023 (Table S2) beyond the MF main effect observed in both years, and reached its maximum normalized value in P3M3-treated fruits only in the second year (Table 4), possibly reflecting cumulative effects of repeated microbial inoculation or enhanced nutrient mobilization under moderate drought. Overall, the contrasting rainfall conditions did not appear to weaken the treatment effects but instead suggest that SAP–MF co-application may confer its greatest relative benefit under water-scarce conditions.

Beyond these soil- and root-level physiological gains, transcriptomic profiling indicated that root nitrogen metabolism was substantially altered under P3M3 at the molecular level. The upregulation of MdGS1, MdGS2, MdGLUD1 and MdNADH points to enhanced ammonium assimilation via the GS and GDH routes, channeling inorganic nitrogen into glutamine and glutamate as precursors for downstream amino acid biosynthesis. Consistently, free amino acids accumulated in the roots, with proline and arginine showing the sharpest increases—metabolites that serve as nitrogen-rich reservoirs and, in water-limited orchards, as osmoprotectants and stress-responsive compounds (Cheng et al., 2023). This root-level reprogramming paralleled a greater nitrogen supply from the soil, which was in turn associated with a reshaped bacterial community. Indeed, the co-application of SAP and MF

reshaped the soil bacterial community without eroding its overall diversity. In the rhizosphere, the treatment drove a marked compositional shift, with Actinobacteriota expanding at the expense of Proteobacteria and Gemmatimonadota. This enrichment is agronomically meaningful, as many Actinobacteria act as plant growth-promoting rhizobacteria and contribute to nitrogen cycling (Wei et al., 2024). Notably, LEfSe identified two recognized diazotrophs enriched under treatment, *Frankia* and *Bradyrhizobium*, both capable of enhancing plant nitrogen acquisition (Zhong et al., 2024). Importantly, neither *Frankia* nor *Bradyrhizobium* is a constituent of the applied MF product, whose microbial components are dominated by *Bacillus* spp. and *Trichoderma harzianum*. This distinction indicates that the observed diazotroph enrichment reflects indirect community restructuring driven by the altered soil environment rather than a direct inoculation effect. This taxonomic signature was mirrored functionally, as FAPROTAX predicted an increase in the nitrogen-fixation pathway, converging with the elevated soil available nitrogen and the upregulated root nitrogen-assimilation genes to reinforce a coherent nitrogen-centered response. In parallel, the predicted potential for xylanolysis and chitinolysis declined under treatment. Because these pathways govern the breakdown of recalcitrant plant- and fungal-derived polymers, their attenuation is consistent with the higher soil organic matter we observed, as elevated nitrogen availability commonly slows microbial decomposition and favors fast-growing copiotrophic taxa over decomposer-dominated guilds (Ding et al., 2024; Fierer et al., 2007). Together, these shifts indicate that SAP–MF co-application coordinates root nitrogen assimilation with a rhizosphere microbiome oriented toward nitrogen enrichment and organic-matter conservation.

In addition to the upregulation of nitrogen metabolism genes, the transcriptomic analysis also revealed significant changes among the downregulated DEGs. KEGG enrichment of the 864 downregulated genes showed that they were predominantly enriched in secondary metabolism pathways, including sesquiterpenoid and triterpenoid biosynthesis, phenylpropanoid biosynthesis, flavonoid biosynthesis, and glutathione metabolism, along with the citrate cycle (Fig. S8). These pathways are commonly activated under drought or nutrient deficiency as part of the plant's stress defense response (Cheng et al., 2023; Li et al., 2023). Their attenuation under P3M3 is consistent with the alleviated abiotic stress conditions resulting from improved soil water and nitrogen status, suggesting a metabolic reallocation from stress-induced secondary metabolism toward growth-oriented primary metabolism in the roots.

Based on the comprehensive analysis of 11 indicators from the two-year experiment, the P3M3 treatment demonstrated the optimal performance in terms of yield, fruit quality and WUE. Moreover, we propose a model depicting how the combination of SAP and MF regulates apple development (Fig. 9). In brief, co-application of SAP and MF improved soil properties and restructured the rhizosphere bacterial community, alongside upregulated root nitrogen metabolism genes (*MdGLUD1*, *MdNADH*, *MdGS1* and *MdGS2*), elevated free amino acid contents, and enhanced apple yield and fruit quality. Two-way ANOVA across 31 indicators over both years further revealed that SAP and MF contributed to apple productivity through largely independent, complementary mechanisms: SAP governed soil moisture dynamics, photosynthetic performance, and fruit enlargement, whereas MF primarily drove nitrogen availability and soluble solids accumulation (Table S2). A statistically significant SAP $\times$ MF interaction was confined to a small subset of indicators, most notably WUE, which showed consistent synergy in both years. Thus, the term 'synergistic' is applied specifically to WUE in this study, while the overall co-application benefit is best characterized as functionally complementary.

Collectively, these findings provide an integrated mechanistic framework for understanding soil–bacteria–root interactions in arid and semi-arid orchards, and offer practical guidance for the sustainable development of fruit production in water-limited regions.

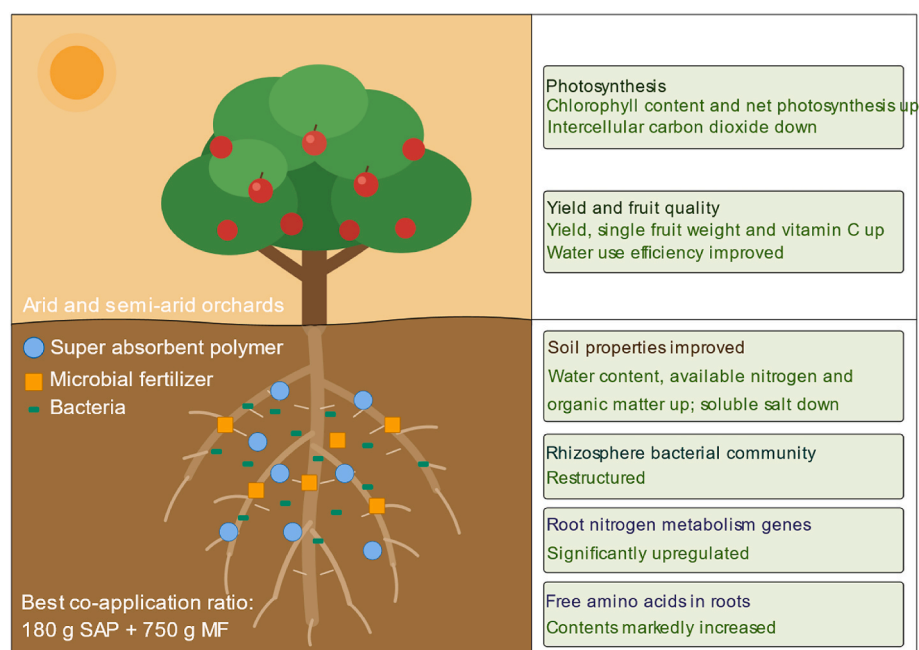


Fig. 9. A proposed working model illustrating the synergistic effects of SAP and MF co-application on apple productivity. Based on a comprehensive evaluation of 11 quality and yield indicators over two years, the P3M3 treatment (180 g SAP + 750 g MF per plant) was identified as the optimal combination. This treatment improves soil water content, available nitrogen, and organic matter while reducing soluble salts, and restructures the rhizosphere bacterial community. These changes upregulate root nitrogen metabolism genes, increasing root free amino acids, which collectively enhance photosynthesis, water use efficiency, yield, and fruit quality.

5. Conclusions

Our results demonstrated that the combined application of super-absorbent polymer (SAP) and microbial fertilizer (MF) improved the soil environment of dryland apple orchards in northern Shaanxi, increasing soil water content and available nitrogen while keeping other soil chemical properties largely stable. These improvements enhanced leaf chlorophyll content and photosynthetic performance, and translated into higher yield, single-fruit weight, water-use efficiency, and fruit quality across both the 2022 and 2023 seasons. A comprehensive evaluation of fruit quality and water-use efficiency consistently identified P3M3 (180 g SAP + 750 g MF per plant) as the optimal treatment in both years. Root transcriptomic profiling of P3M3 relative to the control revealed differentially expressed genes that were significantly enriched in nitrogen metabolism and arginine biosynthesis. Consistently, key nitrogen-assimilation genes (*MdNADH*, *MdGS1*, *MdGS2*, and *MdGLUD1*) were up-regulated, and the contents of free amino acids, including arginine and proline, were markedly increased in roots. Furthermore, 16S rRNA sequencing showed that P3M3 restructured the rhizosphere and bulk-soil bacterial communities without reducing their diversity, enriching Actinobacteriota together with the diazotrophic genera *Frankia* and *Bradyrhizobium*, and increasing the predicted potential for nitrogen fixation. Together, these results demonstrate that SAP combined with MF improves water- and nitrogen-use efficiency, accompanied by coordinated changes in the soil bacterial community, root nitrogen metabolism, and tree physiology, collectively contributing to enhanced apple productivity and quality under water-limited conditions.

CRediT authorship contribution statement

Jieqiang He: Formal analysis, Data curation and Writing – original draft. Xinyue Yang: Investigation. Jiale Wen: Formal analysis. Jiayue Rao: Investigation. Chengkui Ji: Investigation. Yipeng Zhang: Investigation. Jinghao Jiang: Investigation. Runze Liu: Formal analysis. Qingmei Guan: Supervision and Funding acquisition. Fengwang Ma:

Supervision and Funding acquisition. Chana Bao: Investigation. Yutian Zhang: Data curation and Writing – review and editing. Chundong Niu: Supervision and Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.plaphy.2026.111729>.

Data availability

Data will be made available on request.

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